



Soft Tissue Sarcoma Clinical Analytics Dashboard

A Power BI Dashboard for Clinical Decision Support

Presented By

Mfoniso Isaac Udo

Certified Oncology Nurse | Data Analyst

Tools Used

Microsoft Power BI | Power Query

DAX | Excel



Project Overview

Background

Soft tissue sarcoma comprises a diverse group of rare cancers arising from connective tissues such as muscle, fat, nerves and blood vessels. Because of their rarity and complexity, understanding patient demographics, treatment patterns, metastasis and survival outcomes is essential for improving clinical decision-making.

Project Objective

This project aims to transform raw clinical data into an interactive Power BI dash that enables healthcare professionals to:

- Understand patient demographics
- Explore tumor characteristics
- Analyze metastasis and recurrence
- Evaluate treatment approaches
- Assess survival outcomes





Dataset Overview

Dataset Description

Dataset Summary :

- Total Patients: 51
- Clinical Variables: 30+
- Histological Types: 30
- Sarcoma Families: 9
- Treatment Categories: 3
- **Outcome Categories:**
 - No Event
 - Metastasis
 - Recurrence

Data Preparation

The dataset underwent:

- Data cleaning
- Missing value handling
- Standardization
- Feature engineering
- Relationship modelling
- DAX measure creation





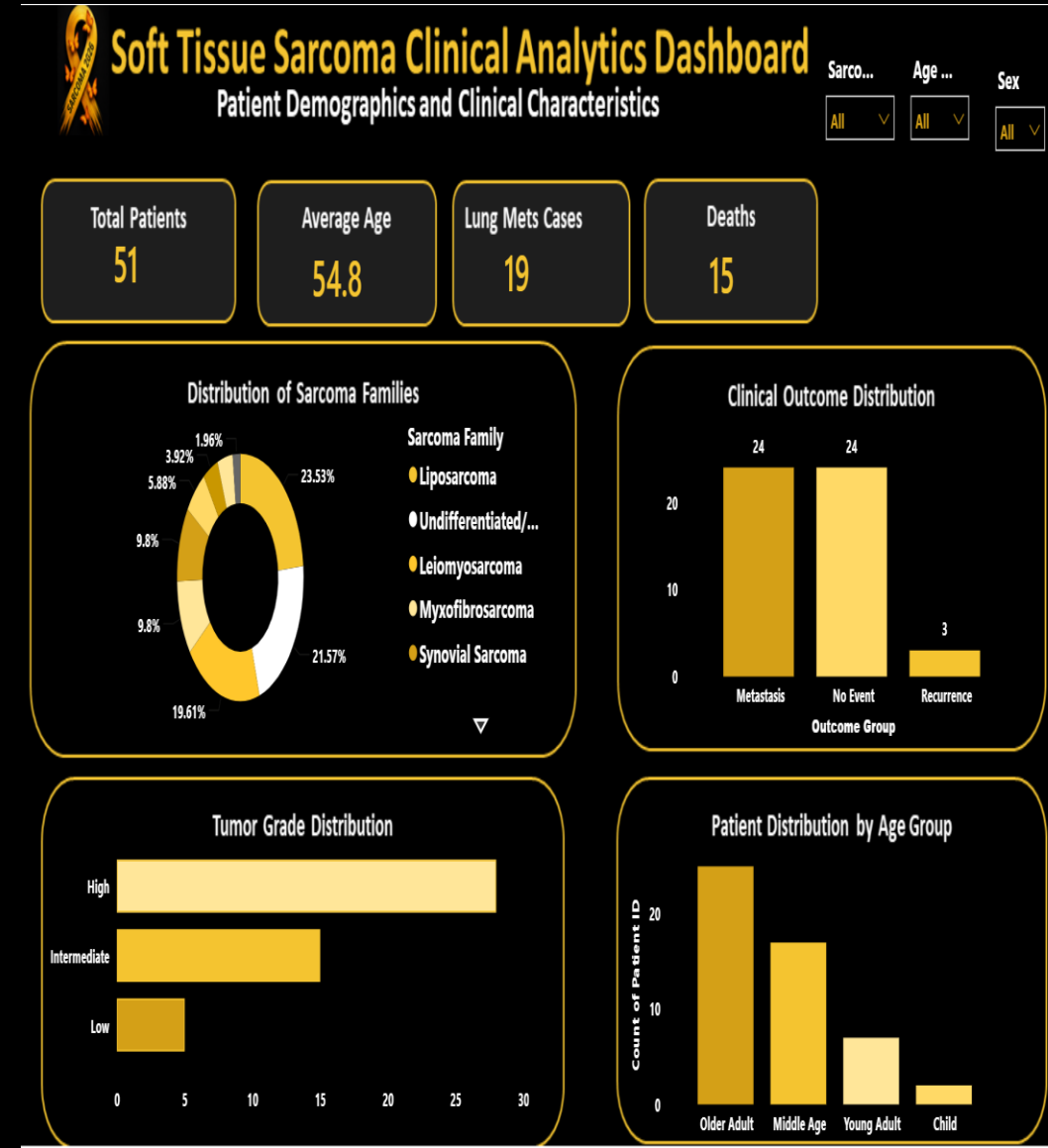
Dashboard Overview

Executive Summary

The dashboard provides a high-level overview of the clinical characteristics of **51 Soft Tissue Sarcoma patients**, highlighting patient demographics, disease burden, tumor classification, and clinical outcomes.

Key Highlights

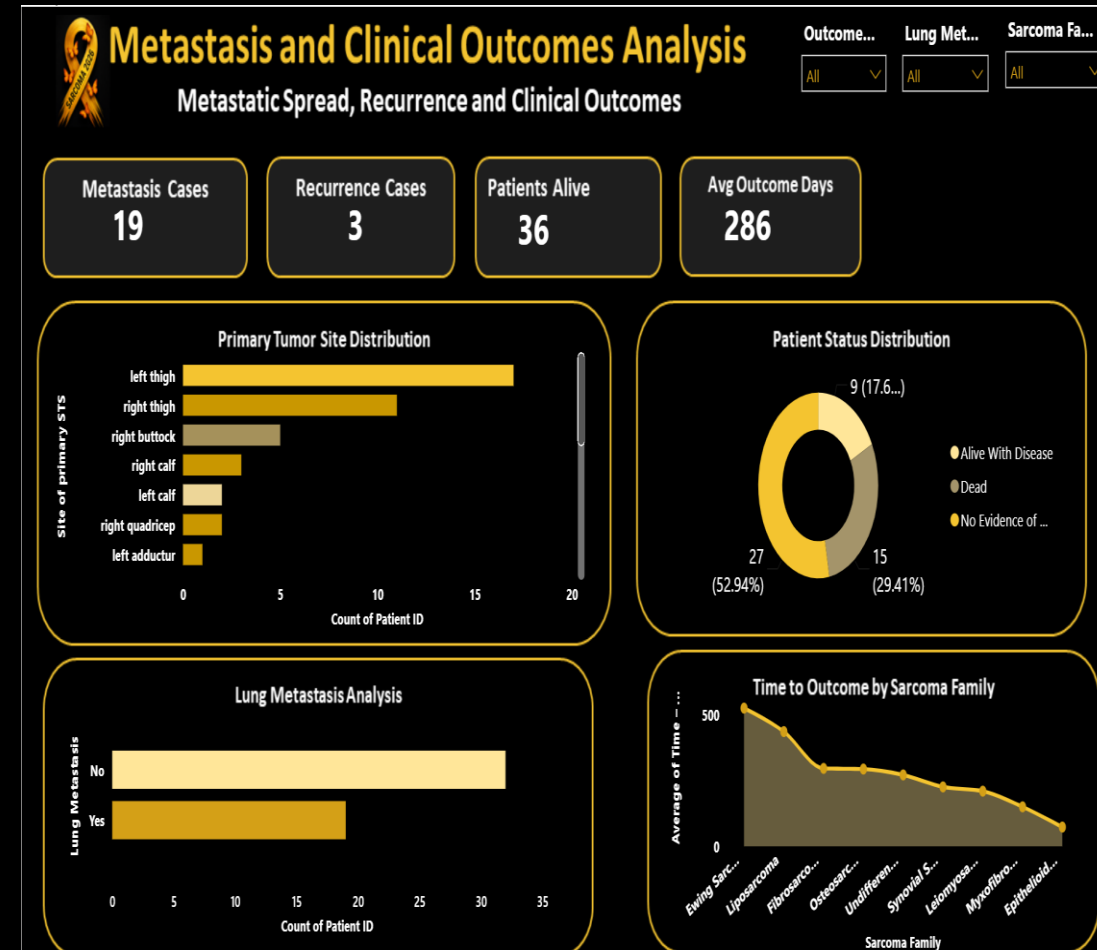
- **51 patients** were included in the analysis.
- The **average patient age was 54.8 years**, with **Older Adults** representing the largest age group.
- **19 patients developed lung metastasis**, indicating a substantial metastatic burden.
- **15 deaths** were recorded during the follow-up period.
- **Liposarcoma** was the most frequently diagnosed sarcoma family.
- **High-grade tumors** accounted for the largest proportion of cases.
- Clinical outcomes were predominantly **No Event** and **Metastasis**, while **Recurrence** was relatively uncommon.



Metastasis & Clinical Outcomes

Key Findings :

- **19** patients developed lung metastasis, making it the most common metastatic event.
- Recurrence was uncommon, with only 3 recorded cases.
- Most patients achieved No Evidence of Disease (NED) following treatment.
- Time to outcome differed across sarcoma families, suggesting variation in disease progression and prognosis.
- Primary tumor sites were concentrated in the lower limbs, particularly the thigh region.



Treatment Patterns

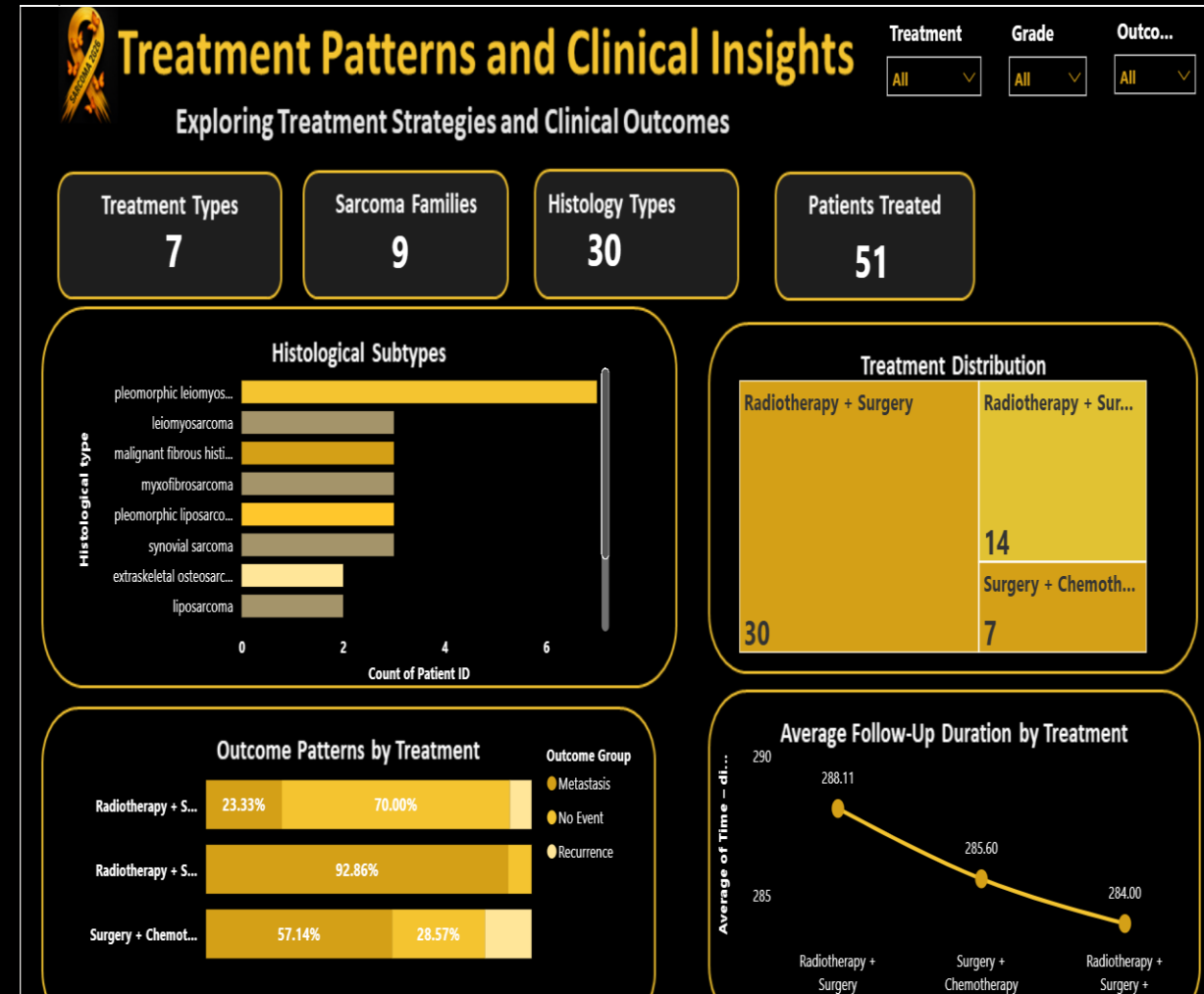
Insight :

The dashboard compares:

- Histological subtypes
- Treatment distribution
- Clinical outcomes by treatment
- Average follow-up duration

Observation :

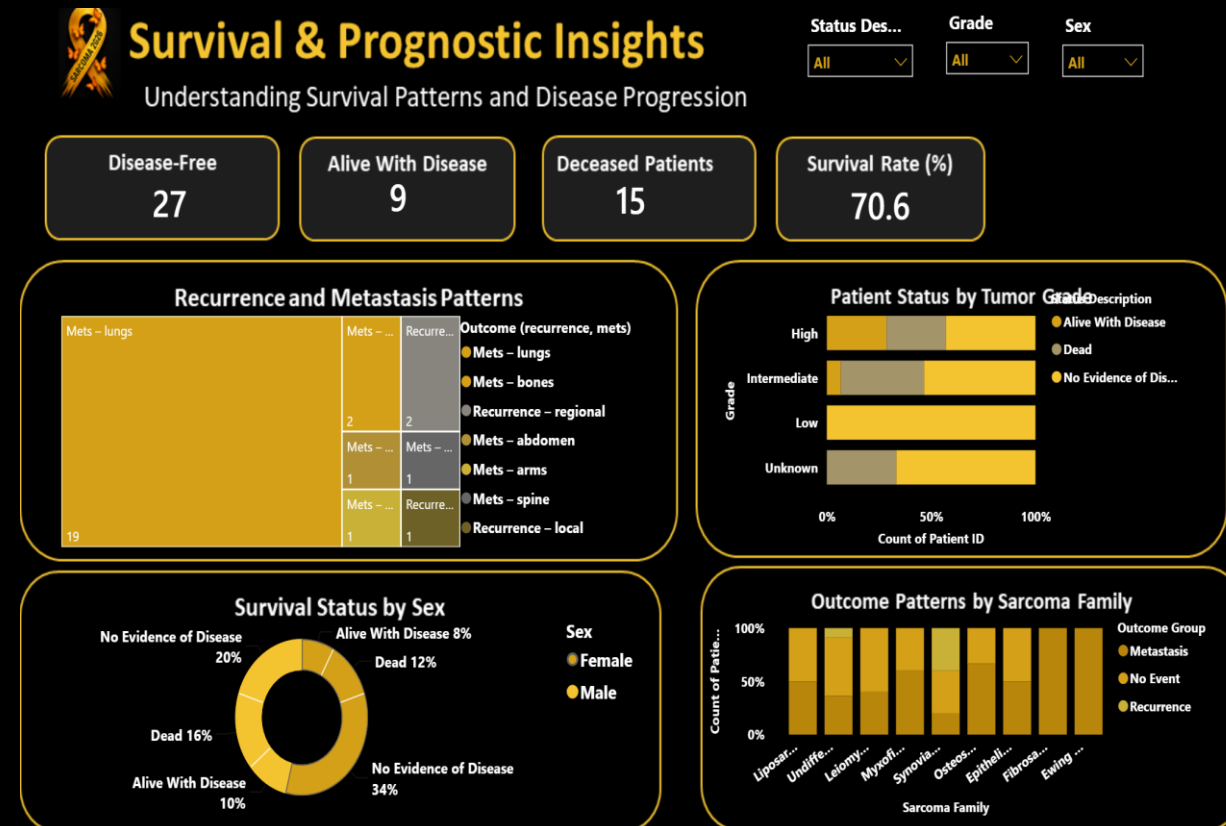
Radiotherapy combined with surgery represented the most common treatment strategy in the dataset.



Survival & Prognostic Insights

Findings :

- 70.6% of patients remained disease-free during follow-up.
- High-grade tumors were associated with less favorable patient outcomes than lower-grade tumors.
- Survival outcomes varied across sarcoma families, highlighting differences in disease behavior.
- Recurrence remained relatively uncommon compared with metastatic progression.
- Survival patterns were comparable across male and female patients in this dataset.





Key Business/Clinical Insights

Major Findings :

- Liposarcoma represented the largest sarcoma family.
- High-grade tumors were more common than low-grade tumors.
- Older adults accounted for the majority of patients.
- Lung metastasis was the predominant metastatic presentation.
- Most patients achieved **No Evidence of Disease** following treatment.
- Radiotherapy with surgery was the most frequently administered treatment.
- Survival outcomes varied across sarcoma families.



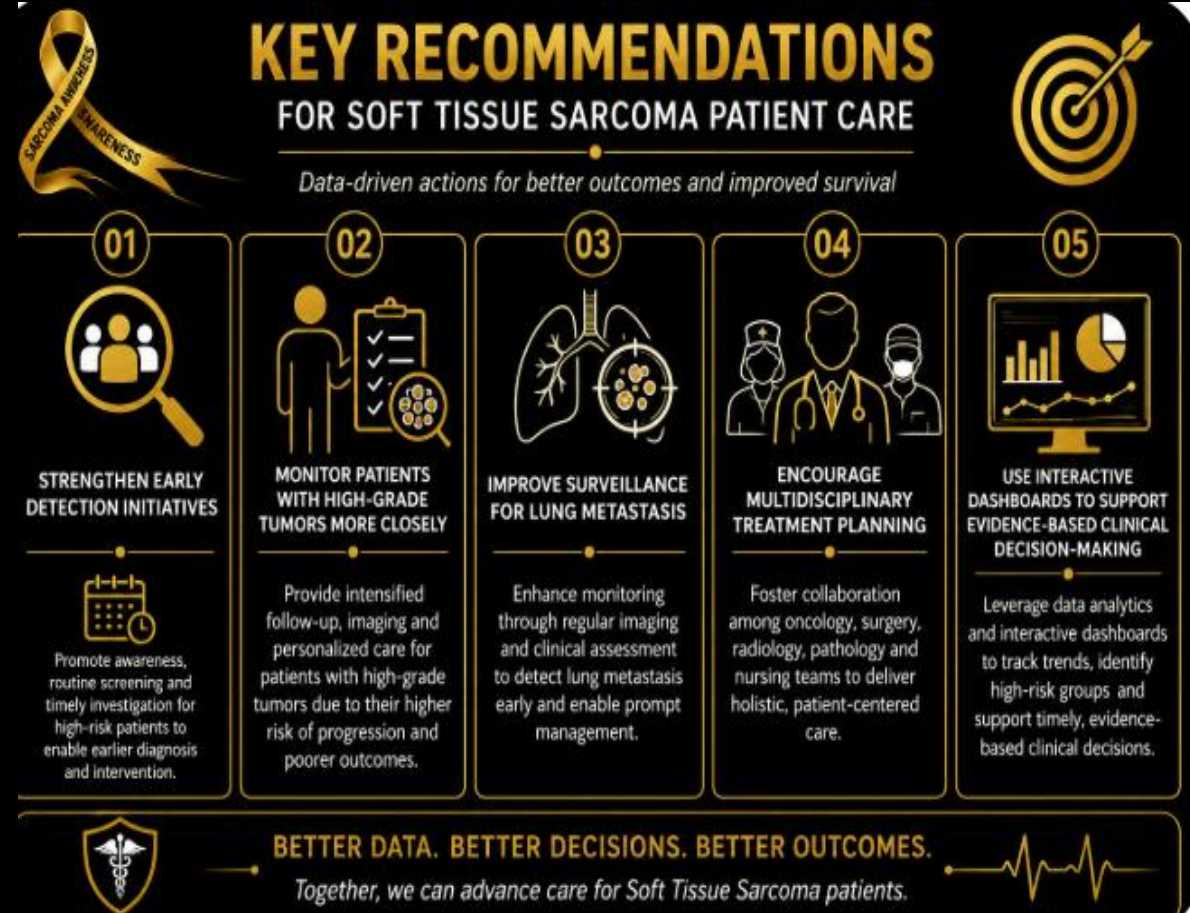


Recommendations

Clinical Recommendations :

•Based on the findings:

- Strengthen early detection initiatives for high-risk patients.
- Monitor patients with high-grade tumors more closely.
- Improve surveillance for lung metastasis.
- Encourage multidisciplinary treatment planning.
- Use interactive dashboards to support evidence-based clinical decision-making.



Conclusion

Clinical Recommendations :

- **Based on the findings:**

- This dashboard demonstrates how healthcare data can be transformed into actionable clinical insights using Power BI.
- Through interactive visualizations, clinicians and healthcare administrators can better understand patient demographics, treatment patterns, metastatic progression, and survival outcomes to support evidence-based decision-making.





THANK YOU

Thank you for your time and attention.



QUESTIONS & DISCUSSION

PRESENTED BY



Mfoniso Isaac Udo

Certified Oncology Nurse | Data Analyst



Together, data-driven insights can contribute to improved clinical decision-making and better patient outcomes.